

UPSHUR COUNTY REGIONAL, WV, USA

WMO: 720328

Lat: 39.000N

Lon: 80.274W

Elev: 1635

StdP: 13.85

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63832

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	10.2	-3.8	4.8	9.1	0.9	6.1	14.2	24.2	42.8	20.9	44.2	5.1	280	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	88.1	72.8	85.7	72.0	83.6	70.7	75.6	83.1	74.5	81.3	73.2	79.5	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	131.3	77.8	72.5	127.9	77.2	71.9	125.4	76.8	40.3	83.4	39.0	82.0	38.2	79.8	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.0	15.0	DB	0.4	91.0	5.6	1.7	-3.6	92.3	-6.9	93.3	-10.1	94.3	-14.2	95.5	
			WB	-0.5	78.1	5.4	2.4	-4.5	79.8	-7.6	81.2	-10.7	82.6	-14.6	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	32.6	34.2	43.0	54.0	62.5	69.0	72.5	71.4	66.7	54.9	44.4	37.9
	DBStd	16.94	13.06	12.37	11.46	9.88	8.18	5.38	4.38	4.67	6.85	9.31	9.80	10.55
	HDD50	2052	558	460	282	71	7	0	0	0	0	55	221	398
	HDD65	4937	1006	863	683	345	148	25	4	7	62	334	620	840
	CDD50	3398	17	17	65	192	394	570	697	662	501	207	53	23
	CDD65	806	0	0	2	16	70	145	236	204	112	20	1	0
	CDH74	5959	0	2	14	177	583	1034	1656	1429	893	165	6	0
	CDH80	1497	0	0	0	28	110	241	461	378	246	33	0	0
Wind	WSAvg	4.9	6.7	6.8	6.4	6.5	4.5	4.0	3.3	2.8	2.9	4.3	5.2	6.0
Precipitation	PrecAvg	46.9	3.3	2.9	4.3	3.9	4.6	4.4	4.8	4.0	3.9	3.2	3.7	3.6
	PrecMax	64.8	5.9	6.8	9.3	6.9	12.8	10.1	9.3	8.2	8.2	8.1	11.6	8.0
	PrecMin	33.9	1.1	0.9	1.4	1.1	1.4	1.2	1.7	1.7	0.4	0.4	1.3	0.9
	PrecStd	6.6	1.4	1.2	1.9	1.5	2.4	2.0	1.8	1.5	1.6	1.7	1.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.8	70.3	75.3	83.7	86.2	88.5	90.5	90.2	89.9	83.9	73.1	69.8
		MCWB	56.4	59.8	58.5	59.7	69.3	71.4	74.9	74.0	70.5	69.9	59.0	60.9
	2%	DB	61.4	63.4	70.4	79.2	82.3	85.7	87.9	87.5	85.8	78.9	68.1	63.3
		MCWB	53.9	54.0	56.3	59.7	67.2	71.9	74.7	73.4	70.7	64.9	56.3	55.1
	5%	DB	56.8	57.5	65.7	74.8	80.9	82.3	84.3	84.0	82.2	73.5	63.8	58.9
		MCWB	51.2	50.0	54.2	57.6	66.5	70.3	72.7	71.8	68.9	62.3	53.7	53.1
	10%	DB	53.7	54.2	61.4	71.7	77.2	80.8	82.0	81.7	79.4	70.2	60.8	54.8
		MCWB	49.9	48.9	51.3	55.8	64.4	69.4	71.5	70.8	67.0	62.0	51.9	50.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	60.7	61.2	65.0	73.2	75.5	78.1	77.0	75.1	72.3	62.7	61.1
		MCDB	63.3	66.1	71.3	76.6	81.0	82.8	85.7	84.8	83.3	77.9	70.8	66.7
	2%	WB	55.2	56.8	58.7	62.7	70.4	74.0	76.1	75.3	73.2	68.9	59.1	57.2
		MCDB	59.6	61.7	66.8	72.2	78.8	81.0	83.7	82.3	80.5	74.7	64.9	61.0
	5%	WB	53.0	53.4	56.6	60.6	68.8	72.5	74.7	73.9	71.5	65.8	56.0	54.3
		MCDB	56.4	57.0	63.0	69.4	76.9	79.5	81.7	80.4	77.6	72.3	61.1	57.7
	10%	WB	49.0	48.1	53.8	58.6	67.0	71.0	73.4	72.6	69.7	62.8	53.2	51.1
		MCDB	53.7	53.2	59.9	67.3	74.4	77.9	79.2	78.6	76.3	67.5	58.9	54.7
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.1	18.1	21.4	19.8	18.5	17.9	18.5	20.1	18.8	17.5	14.9
		MCDBR	19.2	22.4	23.9	26.4	23.9	21.2	20.8	21.4	23.2	24.3	23.2	21.3
		MCWBR	14.2	16.7	14.2	12.5	11.2	9.7	8.8	8.7	9.8	12.2	14.6	15.6
	5% WB	MCDBR	17.6	20.7	20.8	20.8	19.5	17.4	17.8	17.3	18.9	20.3	19.0	20.0
		MCWBR	14.0	17.9	14.0	11.8	9.8	9.0	8.8	8.2	8.9	10.9	14.3	16.2
Clear-Sky Solar Irradiance	taub	0.296	0.314	0.339	0.376	0.419	0.446	0.467	0.482	0.390	0.352	0.316	0.293	
	taud	2.482	2.418	2.418	2.343	2.225	2.185	2.148	2.074	2.328	2.435	2.510	2.525	
	Ebn at Noon	281	289	292	285	274	266	259	251	271	271	271	274	
	Edh at Noon	25	30	34	39	44	46	48	50	36	30	24	22	
All-Sky Solar Radiation	RadAvg	542	789	1128	1491	1679	1858	1781	1654	1379	982	686	478	
	RadStd	62	117	86	124	142	126	105	89	152	120	83	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+106

Nomenclature: See separate page